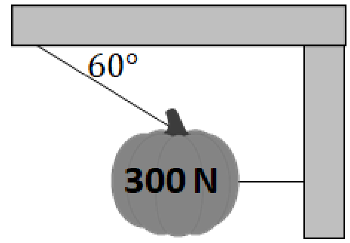
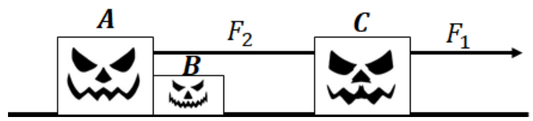
Directions: **You must show all steps** required to arrive at the correct answer for the problem below.

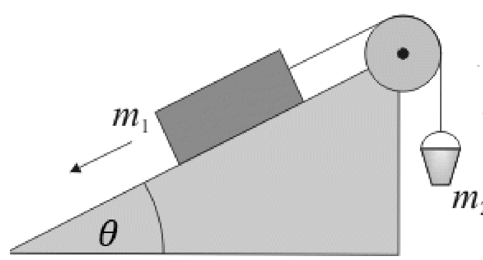
1. A system of two cables supports a 300-N pumpkin as shown.
   a) Calculate the tension in the right-hand (horizontal) cable.
   b) Suppose the angle for the left-hand cable is decreased from 60° to 30°. Would the tension in the left-hand increase, decrease, or remain the same? Justify your answer qualitatively without doing any calculations.



2. 3 blocks move together in the system shown. The blocks, A, B, and C have masses of $m_A = 10\ kg, m_B = 5\ kg, and\ m_C = 10\ kg$ respectively. They are pulled by a force F_1 that has a magnitude of $F_1 = 100\ N$. A rope with a tension of F_2 pulls block A, which then pushes block B forward.
   a) Calculate the value of F_2.
   b) State the direction of the force of block B on block A.
   c) Calculate the magnitude of the force of block B on block A.

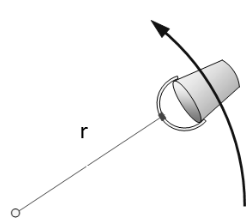


3. The right shows a pulley system. For this system, $m_1 = 80\ kg$, $m_2 = 6kg$, $\theta = 30°$, and the magnitude of sliding friction between m_1 and the incline is $\mu_k = 0.20$. The system is set in motion with m_1 sliding down the incline.



a) Calculate acceleration of the pulley system.
b) Suppose the angle of the incline is increased. How would your answer to b) change? Justify your answer qualitatively without doing any calculations.

c) Suppose the bucket is now taken from the incline and tied to a string of length r = 0.8 m and swung in a vertical circle. The bucket is full of water.
i. Calculate the minimum speed the bucket must have at the top of its circular motion to keep the water in the bucket.
ii. What is the tension in the rope at this point?



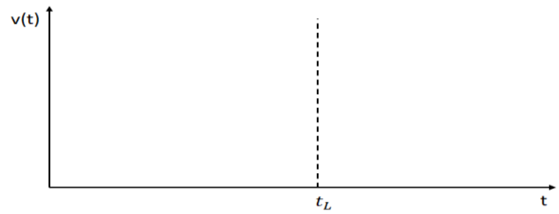
4. An empty sled of mass m = 10.0 kg is placed on one of the many snowy hills of Jupiter, FL. The hill is inclined at an angle of 15° above the horizontal as shown. The coefficient of static friction between the snow and incline is $\mu_s = .40$ and the coefficient of kinetic friction is $\mu_k = .3$.

a) Draw a labeled free body diagram of all forces acting on the sled.

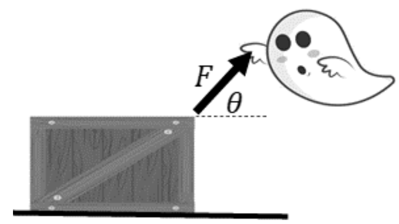
b) The sled rest on the incline without moving. Calculate the force of friction between the sled and incline.

c) The sled is gently nudged and travels down the incline. Calculate the acceleration of the sled down the incline.

d) The sled reaches the bottom of the slope and continues the horizontal ground. Assume the same coefficient of friction. Sketch the velocity of the sled vs. time for the motion. The dashed line at t_L represents when the sled leaves the incline.



5. A ghost pushes a crate of spiders across a rough horizontal surface as shown. He pushes the crate a mass $\boldsymbol{M}$ with a force of magnitude $\boldsymbol{F}$at an angle of $\boldsymbol{\theta}$ above the horizontal.



a) Derive an expression for the coefficient of kinetic friction if the ghost pushes the box forward with a constant velocity.

b) Suppose the ghost pulls the box with a larger angle θ above the horizontal. Would he need to apply more or less force to keep the box moving at constant velocity? Justify your answer qualitatively without doing any calculations.

6. A spooky car drives at 20 m/s around in a circle. The coefficient of friction between the car's tires and the road is 0.45. The car has a spooky mass of 800 kg.

a) Calculate the minimum radius of the circle for the car to remain in circular motion.

b) Calculate the frequency of the car's motion.

c) The car now drives faster than 20 m/s. Briefly describe the subsequent motion of the car.